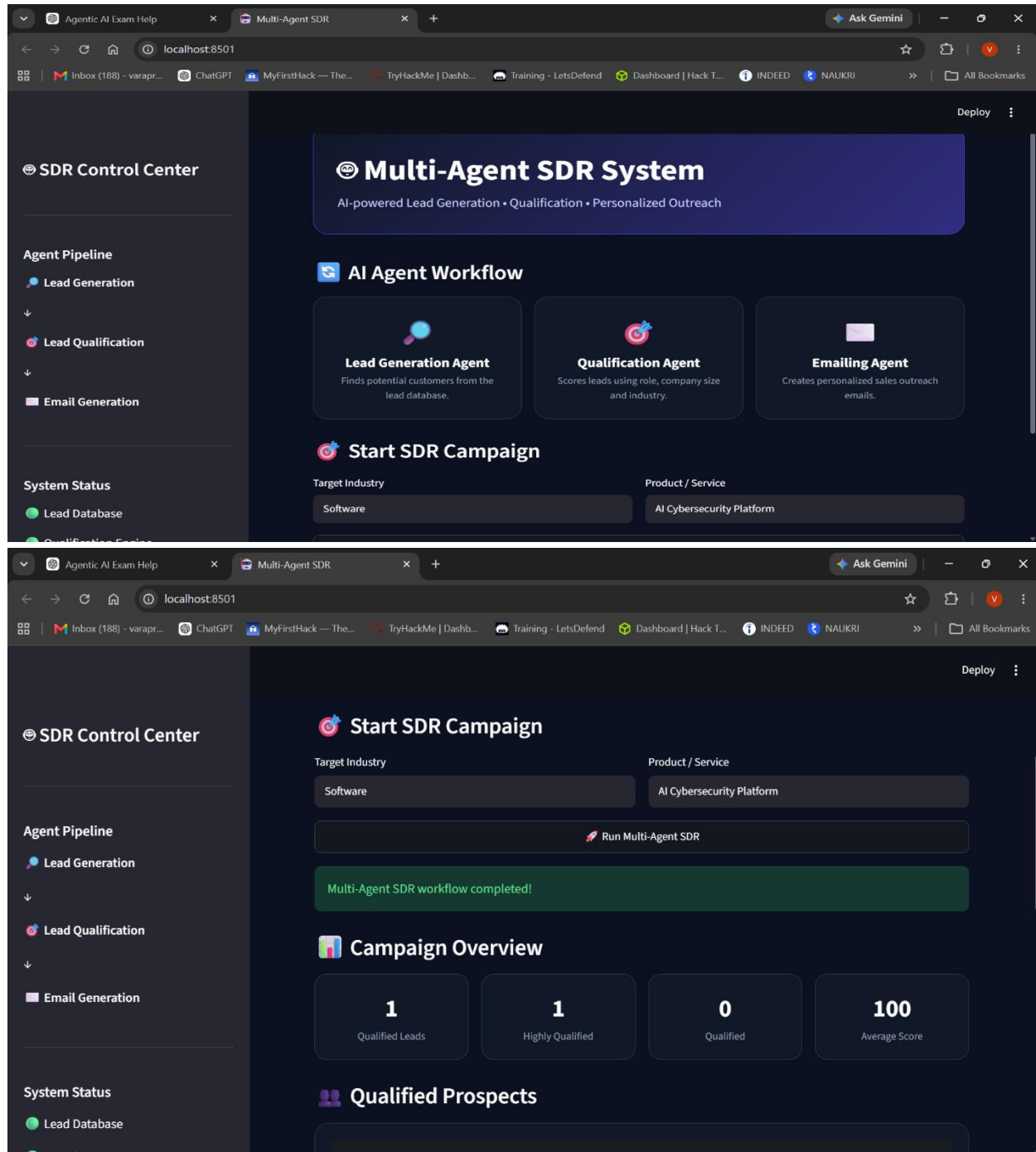


# AGENTIC AI LAB INTERNAL – 1 (21-092026)

1Q. Multi agent SDR system , Design agents for lead generation , qualification and emailing

Output:



Agentic AI Exam Help Multi-Agent SDR Ask Gemini localhost:8501

SDR Control Center

Agent Pipeline

- Lead Generation
- Lead Qualification
- Email Generation

System Status

- Lead Database
- Qualification Engine

### Qualified Prospects

```
<div class="lead-name">
  Rahul Sharma
</div>

<div class="company">
  TechNova Solutions
</div>

<br>

<div class="info">
  <b>Role:</b> CTO
  &nbsp;&nbsp;&nbsp; | &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;
  &nbsp;&nbsp;&nbsp;&nbsp;&nbsp; <b>Industry:</b> Software
</div>

<div class="info">
  <b>Email:</b> rahul@technova.example
  &nbsp;&nbsp;&nbsp; | &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;
  &nbsp;&nbsp;&nbsp;&nbsp;&nbsp; <b>Company Size:</b> 250
</div>
```

Agentic AI Exam Help Multi-Agent SDR Ask Gemini localhost:8501

SDR Control Center

Agent Pipeline

- Lead Generation
- Lead Qualification
- Email Generation

System Status

- Lead Database
- Qualification Engine

### Qualification Score: 100/100

Status: Highly Qualified

> View Personalized Outreach Email

2Q. Text to SQL workflow , Build an end to end LLM with retrieval and query generation

Output:

The image displays two screenshots of a web application titled "AI Text-to-SQL Workflow".

**Top Screenshot:**

- Header:** "AI Text-to-SQL Workflow" with a subtitle: "Convert natural language questions into SQL queries using Retrieval + LLM-style Query Generation + Execution."
- Workflow Steps:**
  - 1. Retrieval:** Retrieve relevant database schema.
  - 2. SQL Generation:** Convert natural language to SQL.
  - 3. Execution:** Execute generated SQL.
  - 4. Result:** Display database answer.
- Ask Your Database:** A section with a text input field. The example question is: "Example: Which product generated the highest revenue?".
- Left Sidebar:** A "Text-to-SQL" section with a "Workflow" list:
  - User Question:** "Natural language input."
  - Retrieval:** "Relevant database schema is retrieved."
  - SQL Generation:** "The question is converted into SQL."

**Bottom Screenshot:**

- Ask Your Database:** The input field now contains "what is highest revenue?". Below it are several example queries in buttons: "What is the total revenue?", "Which product generated ...", "How many customers are ...", "How many orders are there?", "List all products", "Show customers", "Show revenue by product", and "Show orders per customer". A "Run Text-to-SQL Workflow" button is present.
- Success Message:** A green banner states "Workflow completed successfully!".
- Step 1 — Retrieved Schema:** A section titled "orders" showing the schema for the "orders" table, with "order\_id" listed as a column.
- Left Sidebar:** The workflow steps are updated to reflect the current state:
  - User Question:** "Natural language input."
  - Retrieval:** "Relevant database schema is retrieved."
  - SQL Generation:** "The question is converted into SQL."

Text-to-SQL

Workflow

User Question

Natural language input.

↓

Retrieval

Relevant database schema is retrieved.

↓

SQL Generation

The question is converted into SQL.

↓

Step 1 — Retrieved Schema

orders

order\_id

customer\_id

product\_id

quantity

total\_amount

order\_date

Text-to-SQL

Workflow

User Question

Natural language input.

↓

Retrieval

Relevant database schema is retrieved.

↓

SQL Generation

The question is converted into SQL.

↓

Step 2 — Generated SQL

```
SELECT
  p.product_name,
  SUM(o.total_amount) AS revenue
FROM orders o
JOIN products p
  ON o.product_id = p.product_id
GROUP BY p.product_id, p.product_name
ORDER BY revenue DESC
LIMIT 1;
```

Step 3 — Query Execution

SQL query executed successfully.

|   | product_name | revenue |
|---|--------------|---------|
| 0 | Laptop       | 225000  |

Step 4 — Final Result

Agentic AI Exam Help x AI Text-to-SQL x + Ask Gemini

localhost:8501

Text-to-SQL

Workflow

- 1 User Question
- Natural language input.
- 2 Retrieval
- Relevant database schema is retrieved.
- 3 SQL Generation
- The question is converted into SQL.

```
GROUP BY p.product_id, p.product_name
ORDER BY revenue DESC
LIMIT 1;
```

### Step 3 — Query Execution

SQL query executed successfully.

|   | product_name | revenue |
|---|--------------|---------|
| 0 | Laptop       | 225000  |

### Step 4 — Final Result

Rows Returned

1

The natural-language question was converted into SQL and executed against the database.

Q3. RAG based question answering system , implement indexing , retrieval and response generation

Output:

Agentic AI Exam Help x RAG Question Answering x + Ask Gemini

localhost:8501

RAG System

Pipeline

- Document Loading
- Indexing
- Retrieval
- Response Generation

## RAG-Based Question Answering System

Retrieval-Augmented Generation using Indexing • Retrieval • Response Generation

### RAG Pipeline

- 1. Documents**  
Load knowledge from documents.
- 2. Indexing**  
Convert document chunks into searchable vectors.
- 3. Retrieval**  
Find relevant knowledge for the question.
- 4. Response**  
Generate an answer using retrieved context.

Knowledge base indexed successfully • 24 document chunks

Ask a Question

RAG System

Pipeline

Document Loading

↓

Indexing

↓

Retrieval

↓

Response Generation

Ask a Question

Enter your question:

what is rag?

Try an example:

What is Artificial Intelligence?

What is RAG?

What are embeddings?

How does a RAG pipeline work?

What is information retrieval?

What are the advantages of RAG?

Retrieve & Generate Answer

Step 1 — Retrieved Context

> Retrieved Document 1 • Similarity: 1.00

> Retrieved Document 2 • Similarity: 0.49

> Retrieved Document 3 • Similarity: 0.49

Deploy

RAG System

Pipeline

Document Loading

↓

Indexing

↓

Retrieval

↓

Response Generation

Step 2 — Generated Response

Answer

RAG

Workflow Execution

| Indexed Chunks | Retrieved Chunks | Top Similarity |
|----------------|------------------|----------------|
| 24             | 3                | 1.00           |

Deploy